Project Report

Project Title: NeuroRoute AI Edge

Course: Artificial Intelligence (2025–26)

Program & Sem: B.Tech. CE (Div-A), Sem-5

School of Technology Management & Engineering

SVKM'S-NMIMS (Deemed-to-be-University), Indore Campus

[2025-26]

Submitted By	**Submitted To**
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Abstract

This project, NeuroRoute AI Edge, explores the integration of **Swarm Intelligence** and **Federated Learning** to promote sustainable eco-driving in autonomous vehicle fleets. Using a simulated dataset (NeuroRide_Dataset.csv) containing 1000 driving scenarios, the system analyzes parameters such as **average speed**, **driver fatigue**, **terrain strain**, **battery efficiency**, and **carbon emissions**.

Machine learning and graph-based algorithms are used to model fleet-wide optimization through **data-driven insights** and **real-time decision-making**. By combining **Edge AI processing**, **swarm synchronization**, and **eco-efficiency modeling**, this project demonstrates how autonomous systems can collaborate to reduce emissions and improve energy usage.

The research aligns with the **United Nations' Sustainable Development Goal 13 (Climate Action)** and exemplifies how Artificial Intelligence can drive ethical, intelligent, and environmentally responsible automation.

Introduction to Artificial Intelligence — History, Importance, and Scope

Artificial Intelligence (AI) is the field of computer science focused on creating machines capable of **learning, reasoning, and decision-making** similar to human intelligence.

The evolution of AI began with the **Dartmouth Conference (1956)**, where John McCarthy and Marvin Minsky introduced the idea of simulating human thinking. Early AI systems focused on **symbolic reasoning** and **expert systems**, later expanding to **machine learning** and **deep learning** through the 2000s and 2010s.

AI plays a vital role in modern industries by enabling automation, optimizing performance, and improving predictive accuracy. Its scope includes:

- **Perception:** Computer vision and sensor fusion
- **Reasoning:** Logic, inference, and expert systems
- **Learning:** Machine learning, deep learning, and neural networks
- **Interaction:** Natural language processing and robotics

In this project, AI principles are applied to analyze eco-driving behavior, predict carbon impact, and enhance fleet-wide efficiency. By integrating AI into real-world mobility systems, NeuroRoute AI Edge showcases how intelligent decision-making supports **global sustainability** and **smart mobility innovation**.

Project Title

NeuroRoute AI Edge — Swarm Intelligence + Federated Learning for Global Eco-Driving Behavior in Autonomous Fleets.

Overview

The project leverages AI to analyze and optimize driving behavior across autonomous fleets. It introduces a hybrid framework that combines:

- **Swarm Intelligence:** Collaborative vehicle-to-vehicle (V2V) learning.
- **Federated Learning:** Privacy-preserving distributed AI training.
- **Edge AI Processing:** On-device computation for real-time adaptability.

By connecting vehicles as intelligent agents, the system enables **eco-performance optimization**, **battery efficiency management**, and **emission reduction** across fleet networks.

Objective

To develop an AI-driven framework capable of:

1. Predicting eco-driving scores using machine learning.
2. Optimizing vehicle performance via Swarm Sync and Federated AI.
3. Promoting sustainability by minimizing energy waste and carbon output.

Glimpse of Dataset Parameters

The dataset, NeuroRide_Dataset.csv, contains **1000 simulated vehicle trips** across varied terrains, weather, and fatigue conditions.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Vehicle_ID                      1000 non-null   object
 1   Eco_Driving_IQ                  1000 non-null   float64
 2   Avg_Speed_kmph                  1000 non-null   float64
 3   Battery_Usage_Efficiency (%)    1000 non-null   float64
 4   Driver_Fatigue_Level (0-5)      1000 non-null   int64
 5   Terrain_Strain_Index            1000 non-null   float64
 6   Distance_Travelled_km           1000 non-null   float64
 7   Carbon_Emission_gpkm            1000 non-null   float64
 8   Edge_Computing_Decision_Lag_ms  1000 non-null   float64
 9   Swarm_Update_Sync_Flag          1000 non-null   int64
dtypes: float64(7), int64(2), object(1)
memory usage: 78.3+ KB
```

Figure 1: Overview of Dataset Parameters in NeuroRide_Dataset.csv

Each record includes key operational, environmental, and behavioral parameters:

Parameter	Description
Vehicle_ID	Unique identifier for each vehicle
Eco_Driving_IQ	Composite eco-driving performance score
Avg_Speed_kmph	Average vehicle speed
Battery_Usage_Efficiency (%)	Energy efficiency during the trip
Carbon_Emission_gpkm	Carbon emitted per kilometer
Terrain_Strain_Index	Terrain difficulty score
Driver_Fatigue_Level (0–5)	Simulated human-like fatigue level
Edge_Decision_Lag_ms	Edge processing delay
Swarm_Sync_Flag (0/1)	Indicates if vehicle participated in collaborative learning

These parameters together help identify how environment, behavior, and AI synchronization influence eco-driving outcomes.

Data Analysis & Insights

This section focuses on statistical and exploratory data analysis to uncover behavioral patterns and model readiness.

Descriptive Statistics

Using Python's describe() function, we computed the mean, standard deviation, and quartiles of each feature. Most vehicles scored **Eco_Driving_IQ between 65–85**, indicating stable eco-performance.

	Eco_Dr		_Usage_Efficiency (%)	Driver_Fatigue_Level (0–5)	Terrain_Strain_Index	Distance_Travelled_km	Carbon_Emission_gpkm
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	75.168820	51.062580	85.023970	2.600000	1.206130	97.635580	121.024110
std	9.718462	14.961911	4.899351	1.682507	0.193951	100.505202	29.485153
min	42.590000	5.890000	69.900000	0.000000	0.670000	0.060000	50.000000
25%	68.520000	40.902500	81.757500	1.000000	1.080000	27.827500	100.505000
50%	75.255000	50.950000	85.000000	3.000000	1.200000	68.820000	121.110000
75%	81.482500	60.930000	88.307500	4.000000	1.340000	130.482500	140.940000
max	100.000000	97.900000	100.000000	5.000000	1.930000	879.990000	227.980000

Figure 2:Statistical Summary of Fleet Data

Eco Driving IQ Distribution

A histogram shows a near-normal distribution of eco-driving IQ across the fleet.

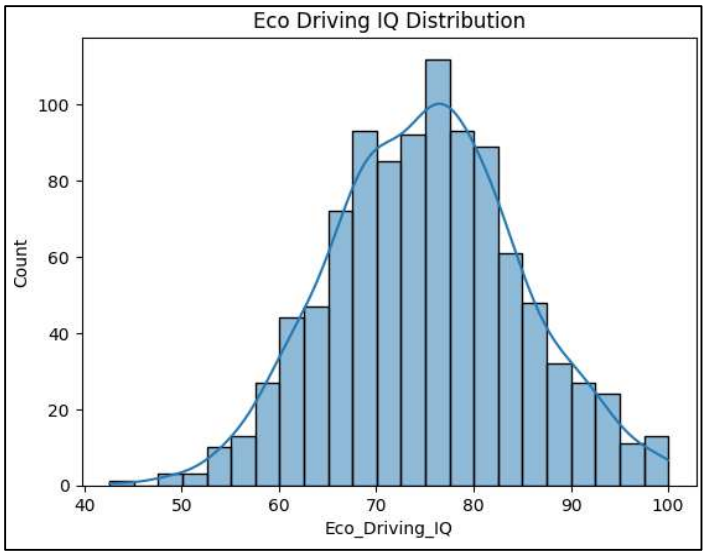

Figure 3: Distribution of Eco Driving IQ across all vehicles

Insights:

- Most scores lie between **70–80**, showing balanced eco-performance.
- Few vehicles below 60 → potential candidates for optimization.
- Supports feasibility for classification and regression models.

Driver Fatigue vs Eco Driving IQ

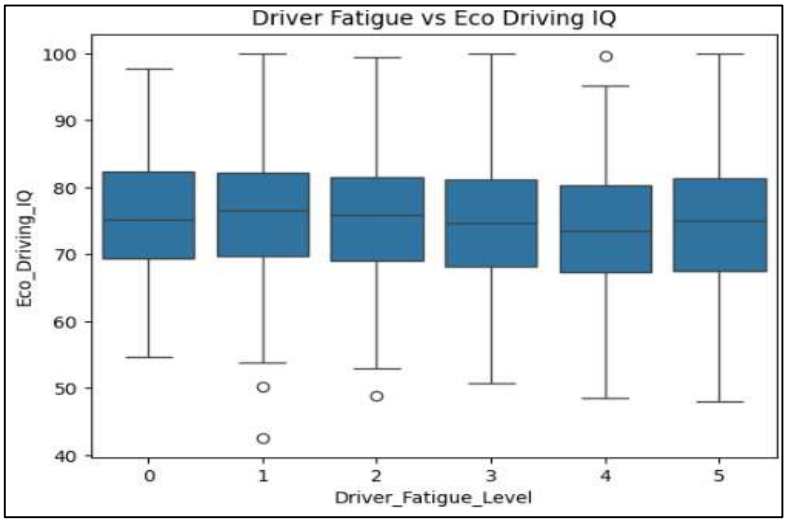

Figure 4: Impact of Driver Fatigue Level on Eco Driving IQ

Boxplot shows how fatigue affects driving efficiency.

Key Insight: Higher fatigue levels increase variability in eco-driving IQ but not necessarily the average score — suggesting fatigue leads to **inconsistent performance**.

Correlation Heatmap

Heatmap of numerical parameters (Pearson correlation).

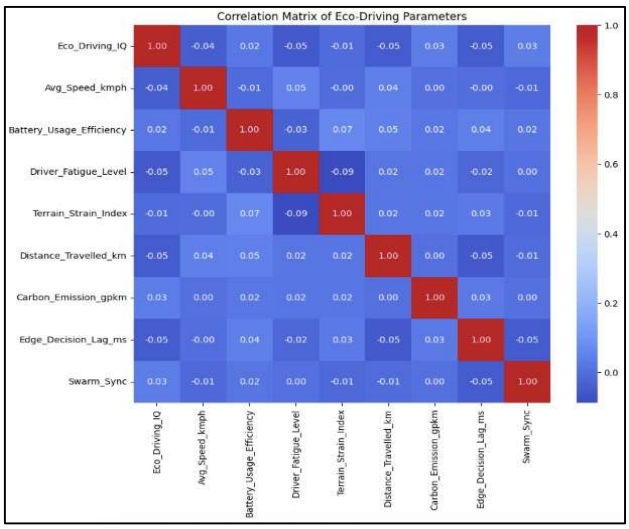

Figure 5: Correlation Matrix between Key Eco-Driving Variables

Findings:

- Weak correlations among features → performance depends on multiple independent factors.
- Slight negative correlation between **Edge Decision Lag** and **Eco IQ**.

Eco Driving IQ vs Carbon Emission

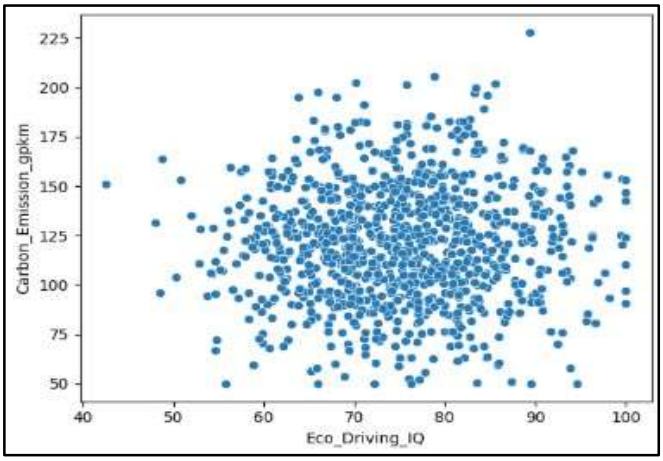

Figure 6: Relationship between Eco Driving IQ and Carbon Emission (g/km)

Scatter plot showing carbon impact relative to eco-IQ.

Insight: No strong correlation, but vehicles with high IQ (>85) tend to emit less carbon — reflecting **smart, efficient driving**.

Terrain Strain vs Battery Usage Efficiency

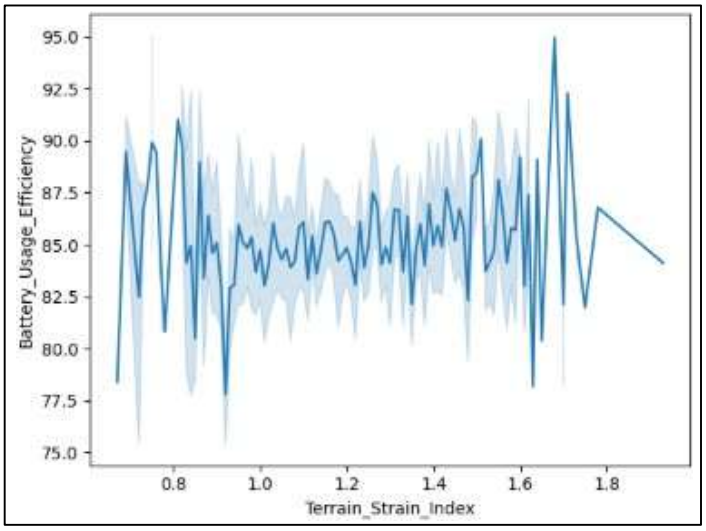

Figure 7: Battery Usage Efficiency vs Terrain Strain Index

As terrain strain increases, battery efficiency slightly decreases — showing **higher energy usage on difficult terrain**.

Driver Fatigue vs Average Speed

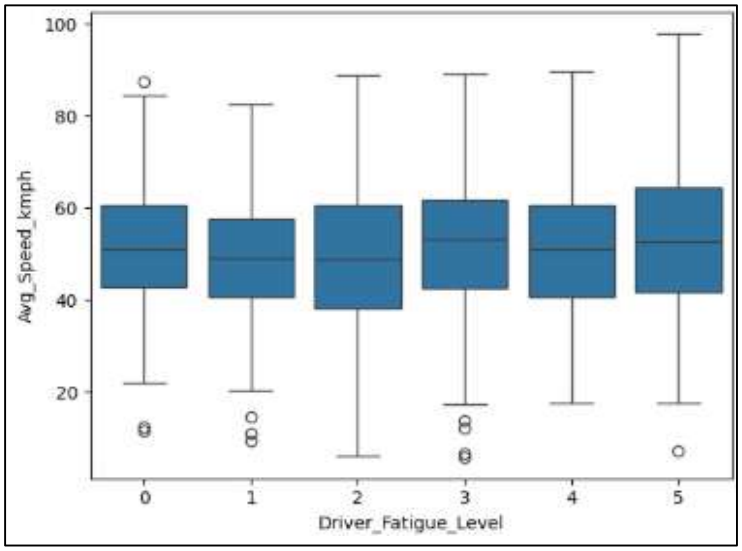

Figure 8: Average Speed vs Driver Fatigue Level

As fatigue rises, **average speed variability increases**, indicating possible risk-taking or inconsistent pacing.

Carbon Emissions vs Distance Travelled

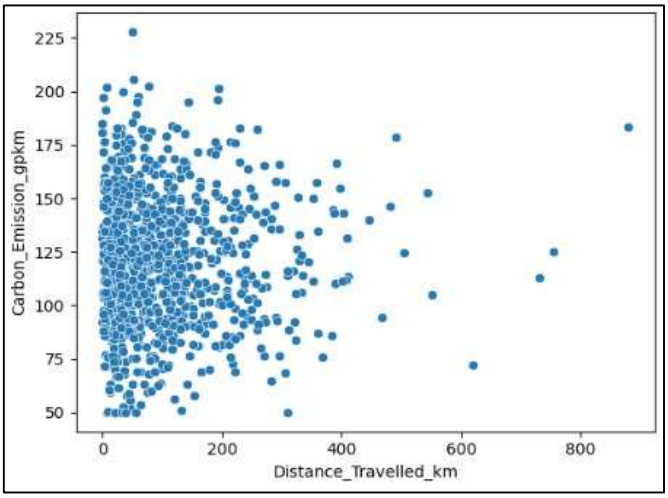

Figure 9: Carbon Emissions vs Distance Travelled

Conclusion: Trip length alone doesn't determine emissions — **driving style and terrain** are more critical.

Machine Learning Approaches in NeuroRoute

Machine learning helps predict eco-performance and optimize fleet decisions.

Regression Analysis

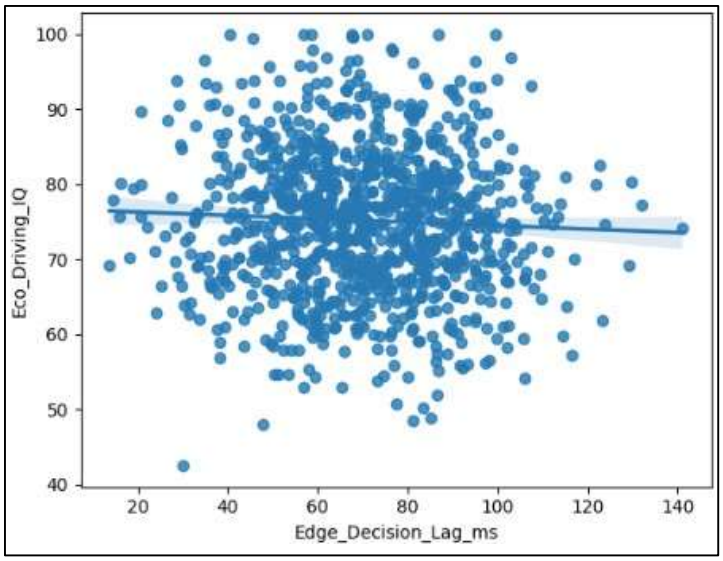

Figure 10: Regression Analysis — Eco Driving IQ vs Edge Decision Lag

Flat regression indicates **latency has minimal effect** on eco-performance, ensuring stability in real-world edge systems.

Clustering (K-Means)

Vehicles are grouped into **Low, Medium, and High Eco Drivers**.
Used for **targeted driver training** and **reward programs**.

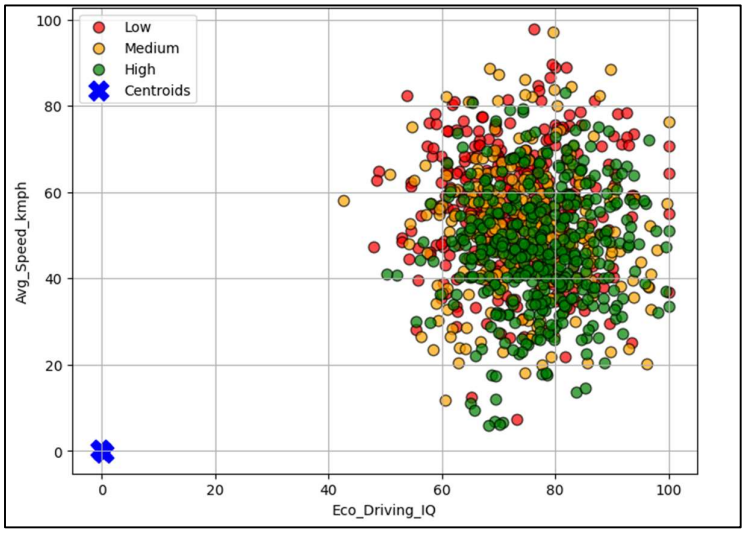

Figure 11: K-Means Clustering of Vehicles by Eco Performance

Swarm Sync Comparison

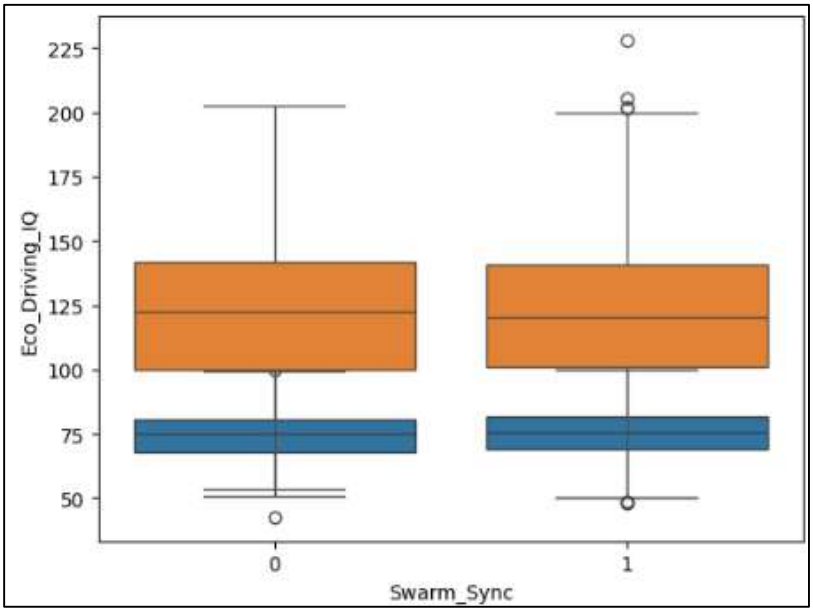

Figure 12: Impact of Swarm Sync on Eco Driving IQ and Carbon Emissions

When Swarm Sync is active:

- Eco IQ ↑ increases
- Carbon Emission ↓ decreases
  → Proves **collaborative AI learning improves sustainability.**

Terrain Strain vs Carbon Emission (Swarm Sync Impact)

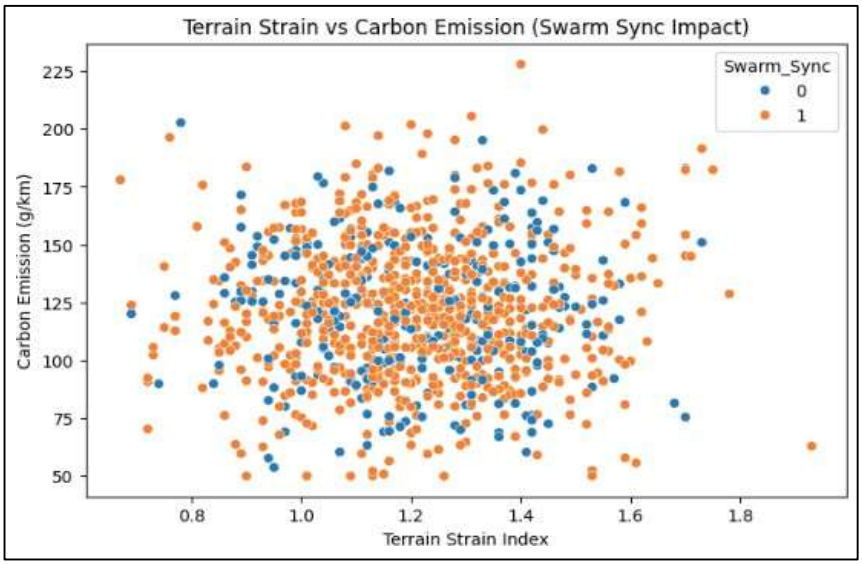

Figure 13: Effect of Terrain and Swarm Sync on Carbon Emission (g/km)

Collaborative learning yields **lower emissions** even in rough terrains.

BFS–DFS Visualization in Fleet Network

Graph algorithms simulate **V2V communication** in the fleet.

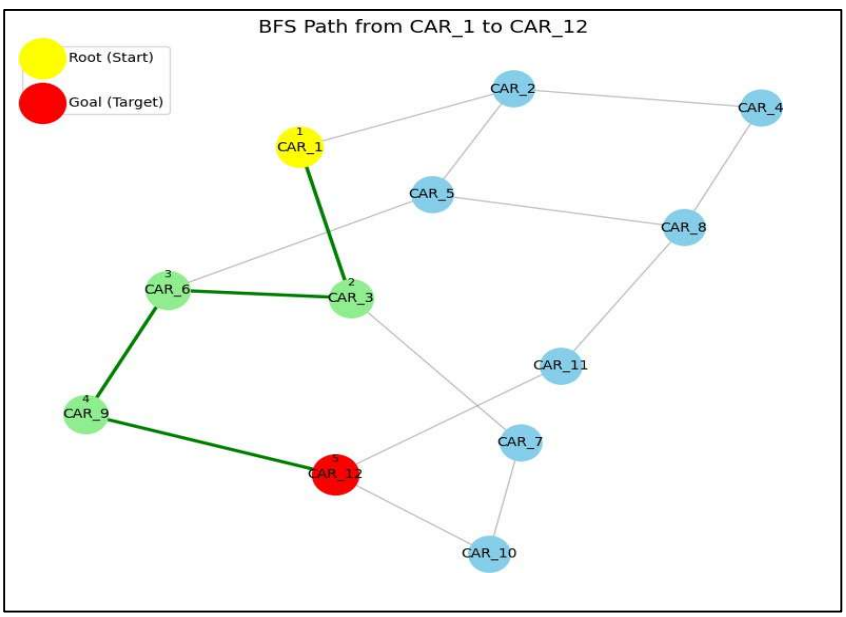

Figure 14: BFS Visualization — Shortest Communication Path between CAR_1 and CAR_12

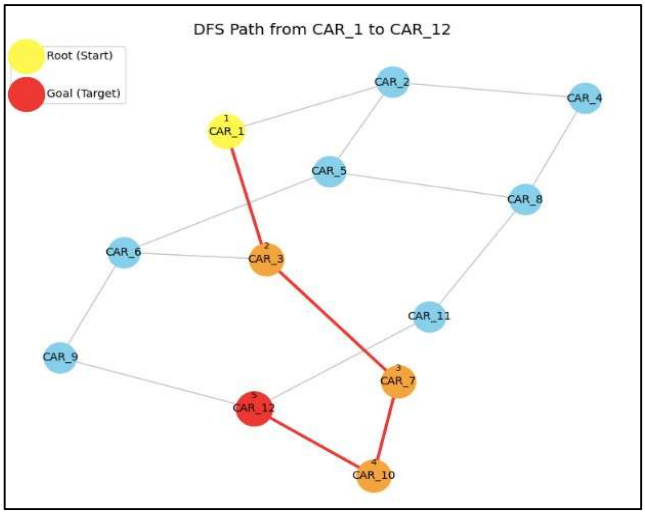

Figure 15: DFS Visualization — Deep Exploration of Fleet Network

Insight:BFS ensures **faster swarm updates**, DFS helps in **network diagnostics and anomaly detection**.

Problem-Solving and Search Strategies

AI search methods enable **optimal decision-making**:

- **Uninformed Search:** BFS, DFS for communication.
- **Informed Search:** A*, Hill Climbing for route optimization.
- **Constraint Satisfaction:** Minimize emissions while maintaining safety.

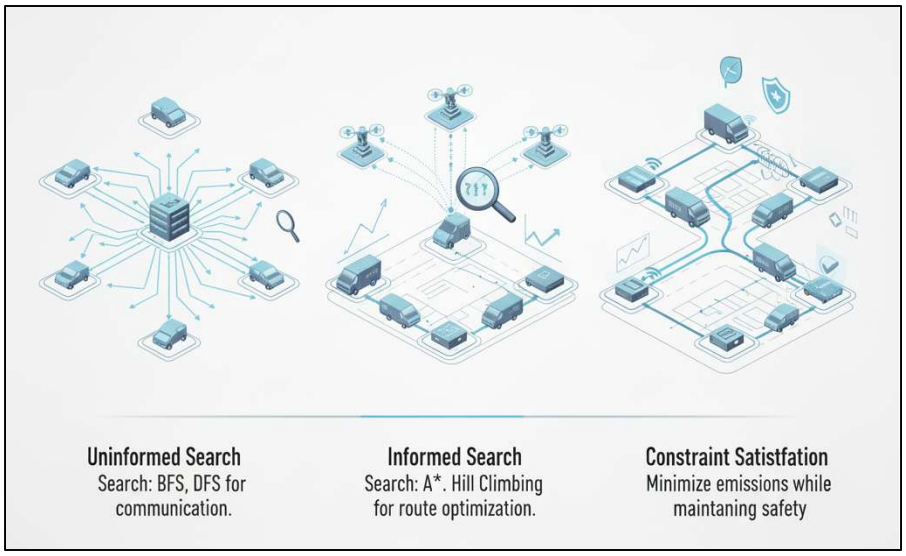

Figure 16: AI Search and Optimization Strategies in Fleet Networks

PEAS and Agent Modeling in NeuroRoute AI Edge

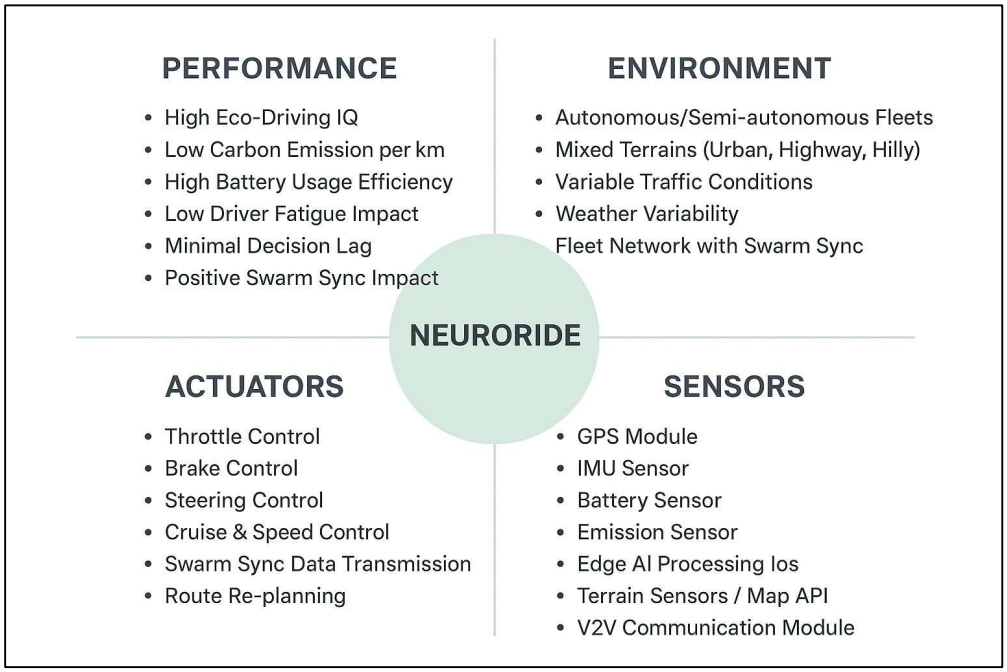

Figure 17: PEAS Model Representation of an Autonomous Vehicle Agent

Each vehicle acts as a **rational agent**, perceiving, deciding, and acting to achieve maximum eco-efficiency.

Knowledge Representation & Expert Systems in Fleet Logic

Rule-based reasoning allows automated decision-making.

Example Rules:

- IF fatigue > 4 AND emissions > threshold → Activate Eco-Mode
- IF terrain strain ↑ THEN → Re-plan route for energy efficiency

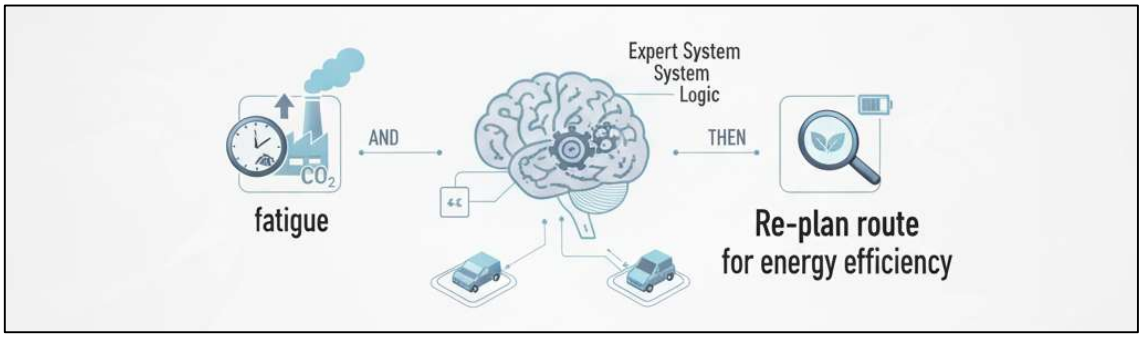

Figure 18: Rule-Based Expert System for Fleet Logic

This hybrid reasoning ensures transparency and adaptability in eco-performance optimization.

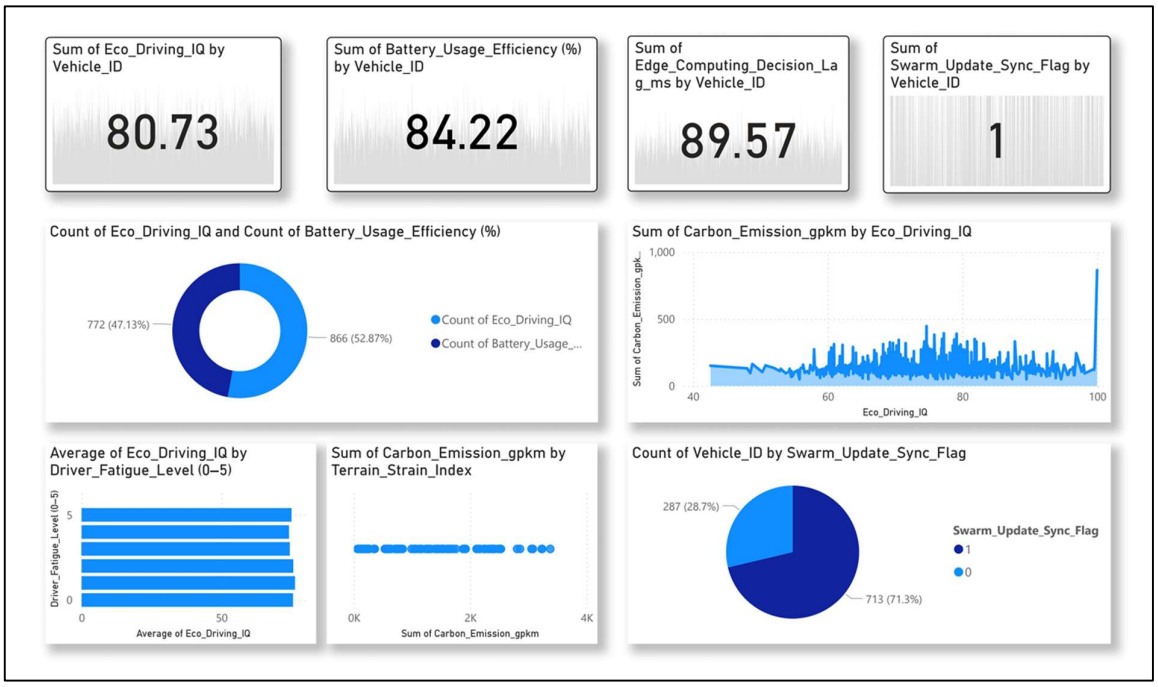

Figure 19:Power BI Dashboard for NueroRouti AI Edge

Real-World Use Cases

- **Fleet Emission Reduction** – Use Eco Driving IQ and Swarm Sync to cut fuel use and CO_2, with real-time eco-driving suggestions.
- **Driver Coaching & Incentives** – Cluster driving styles to target training; reward high performers and provide AI-guided coaching to others.
- **Predictive Maintenance** – Monitor emissions, battery efficiency, and terrain strain to schedule maintenance, reducing downtime.
- **Smart Route Optimization** – Plan routes balancing travel time, energy use, and terrain strain.
- **Edge AI Safety & Efficiency** – Prioritize safety-critical edge decisions without impacting eco-performance.
- **Compliance & Reporting** – Generate automated reports for emissions regulations and sustainability targets.

Ethical, Societal, and Environmental Implications

AI-driven systems must align with ethical and sustainable practices. NeuroRoute adheres to responsible AI principles:

- **Transparency:** Rule-based explanations make decision-making traceable.
- **Fairness:** All vehicles are evaluated using objective performance metrics.
- **Environmental Ethics:** The project supports emission reduction and resource conservation.

- **Social Impact:** Promotes safer, greener logistics, contributing to cleaner urban mobility.

Through these values, NeuroRoute exemplifies AI's role in achieving sustainable innovation responsibly.

Important Findings

- **Eco Driving IQ Stability** – Most vehicles score 65–85 IQ, supporting reliable performance modeling. Fatigue increases variability but not median performance; AI can adapt strategies accordingly.
- **Distance ≠ Efficiency** – Emissions vary widely for the same distance; driver behavior and conditions are more influential.
- **Swarm Sync Benefits** – Collaborative intelligence raises Eco Driving IQ and reduces emissions across terrains.
- **Terrain Effects** – Rough terrain slightly lowers battery efficiency, but emissions remain mostly unaffected.
- **Graph-Based Clustering** – DFS/BFS clusters of driving profiles enable targeted interventions, anomaly detection, and efficient training.

Conclusion

The NeuroRoute AI Edge project successfully integrates **AI principles, machine learning, and swarm intelligence** to create a sustainable, intelligent fleet management system. By applying theoretical foundations such as **PEAS modeling, search strategies, CSPs, and expert systems**, the project demonstrates how AI can enhance real-world transportation efficiency.

Its outcomes show measurable sustainability improvements, data-driven optimization, and responsible AI usage. The project not only fulfills academic objectives but also contributes to the broader goal of **eco-friendly automation and intelligent mobility**—where every vehicle acts as a learning, collaborative, and self-optimizing agent.